

DAVID SHIMIZU

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EDUCATION

University of California San Diego

B.S. Mechanical Engineering, GPA: 3.70/4.00

Fall 2024 – Expected March 2027

La Jolla, CA

El Camino College

Engineering Transfer Coursework

Fall 2022 – Spring 2024

Torrance, CA

TECHNICAL SKILLS

Mechanical Design: SolidWorks, Fusion 360, Tolerance Stack-Up Analysis, GD&T, Enclosure Design

Analysis & Simulation: FEA (ANSYS, SolidWorks Simulation), CFD (SolidWorks Flow Simulation), MATLAB, PID Control

Manufacturing: 3D Printing, Laser Cutting, CNC, PCB Assembly

Embedded & Electronics: Altium Designer (PCB Design), ESP32, C++, Python, ROS 2, BLE, Linux, Git

PROJECTS

Edge Compute Thermal Design | SolidWorks, Flow Simulation

Summer 2026

- Designed and tested a cooling assembly for a Jetson Orin Nano dissipating 21 W under sustained inference, reducing steady-state package temperature from 75°C to 62°C versus the stock cooler at matched power and ambient temperature.
- Swept fin count and spacing across 12 configurations in SolidWorks Flow Simulation using the fan's P-Q curve, selecting 18 fins at 1.5 mm spacing to balance added surface area against rising flow impedance.
- Packaged a heatsink on a CNC-machined 6061 base with a ducted fan path and 3D-printed enclosure, preloading the module to 30 psi with spring-loaded fasteners under NVIDIA's 60 psi limit.
- Increased fin-slot clearance by 0.2 mm per side after supplier DFM review to resolve internal-corner interference.

Autonomous Vehicles | MAE 148 | SolidWorks, ROS2, Linux, AI HAT+

Winter 2026

- Integrated the electromechanical powertrain and sensor suite of a 1/8-scale vehicle on a 4-person team, delivering camera-based path following, RTK GPS waypoint navigation, and vision line following at 20 Hz closed-loop control.
- Designed and 3D-printed a chassis mounting plate and sensor housing in SolidWorks to package the Raspberry Pi 5, OAK-D camera, and VESC, optimizing weight distribution and designing mounts to maintain critical sensor alignment.
- Tuned steering and throttle PID controllers empirically on track, balancing response speed against overshoot and oscillation across successive runs to cut lap time from 145 s to 53 s without sacrificing cornering stability.
- Offloaded vision inference to a Hailo AI HAT+ NPU and eliminated per-frame connection overhead, cutting CPU load from 16% to under 3% and raising standalone model benchmark throughput from 309 to 2,112 FPS.

Dual-Radio Calculator & Bluetooth HID Macropad | Fusion 360, Altium Designer, C++

Summer 2025

- Designed a multi-part calculator enclosure and switch mounting plate in Fusion 360, incorporating heat-set inserts and integrated mounts for a display and LiPo battery.
- Developed a 25-key calculator and HID macropad on ESP32-S3 with Wi-Fi/BLE connectivity, interrupt-driven matrix scanning, and deep sleep during inactivity, measured keypress-to-HID-report latency below 4 ms.
- Specified MJF Nylon PA12 for the enclosure toughness and aesthetics, SLA resin for the keycaps to achieve high feature resolution, and a 1.2 mm FR4 switch plate to provide stiffness during keystrokes.
- Engineered a 4-layer PCB with an onboard LiPo charging circuit and RF keep-out zones to prevent antenna detuning.

Bluetooth 5.0 HID Dongle | Fusion 360, Altium Designer

Fall 2025

- Designed a two-piece press fit enclosure, running a tolerance stack-up to hold fit across 3D-printed parts.
- Designed and validated a 4-layer PCB with impedance-matched RF traces, verified to 280 ft LOS range at -88 dBm RSSI.

EXPERIENCE

NASA Community College Aerospace Scholar

Spring 2023

Mission Design Team

Remote

- Designed a Mars mission architecture with a 10-person team, running trade studies against NASA mass constraints.
- Delivered a Preliminary Design Review to NASA engineers, justifying subsystem selection and mission architecture.

Maui Chicken

Fall 2022 – Summer 2024

Cashier

Torrance, CA